

Atmos

Urban Air Quality & Health-Risk Intelligence

Project explanation · Data Science with Python · September 2026

Problem statement

Cities monitor air pollution with many ground stations, but the raw records are hard to turn into decisions. High pollution days harm health, yet they are difficult to anticipate and to explain. This project builds a complete, honest data science workflow on real government monitoring data: clean it, understand it, test it, model it, and present it.

Main objectives

- Analyse five years of daily air quality and weather for the Los Angeles (South Coast) air basin.
- Identify the conditions under which ground-level ozone exceeds the US health standard of 70 ppb.
- Build simple, well-evaluated models (classification and regression) with strict time-based testing.
- Communicate everything through statistics, clear charts and an interactive dashboard.

Project at a glance

Item	Detail
Data source	US EPA Air Quality System (AQS) daily summary files - official, public-domain monitoring data
Variables	PM2.5, Ozone (O3), NO2 + Temperature and Wind speed (support variables)
Geographic scope	California South Coast air basin: Los Angeles, Orange, Riverside, San Bernardino counties (54 monitors)
Time period	1 January 2021 - 31 December 2025 (1,826 days; 274,467 site-day observations)
Secondary dataset	1,200 US Federal Register EPA documents (for the NLP module)
Integrity rules	Real data only; chronological train/test splits; every number produced by executed code

End-to-end workflow

Data	»	Cleaning	»	Analysis	»	Statistics	»	ML	»	Clustering	»	NLP	»	Forecasting	»	Dashboard
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Each stage is a Python module in the project's `src/` package, so the whole pipeline can be re-run end to end from the raw downloads.

Page 2 - Data & Technologies

What data is used

The US EPA publishes official daily air-quality summaries for every monitor in the USA (AQS "daily" files, one file per pollutant per year). The project downloads 25 such files (5 pollutants x 5 years, about 3.3 GB) and keeps only the four South Coast counties. Everything is real, public and reproducible - no simulated data anywhere.

Main variables (basin-average values, 2021-2025)

Variable	Simple meaning	Unit	Mean
PM2.5	Very fine particles (smaller than 2.5 micrometres) from fuel burning and smoke; small enough to reach deep into the lungs	µg/m³	9.8
Ozone (O3)	Ground-level smog: not emitted directly, it forms when vehicle emissions cook in sunlight - worst on hot, sunny days	ppb	36.3 (daily max is the model target)
NO2	A traffic and combustion gas; a marker for fresh vehicle pollution	ppb	12.9
Temperature	Daily temperature (weather driver: heat speeds up ozone formation)	°F	66.1
Wind speed	Ventilation: wind disperses pollution (slow wind traps it)	knots	3.3

Data cleaning (why it matters)

- Some stations run several instruments for the same pollutant; the project keeps exactly one record per station per day using a documented priority rule (reference method first).
- PM2.5 comes in different "sample durations". These were compared empirically before choosing: two agree within 0.24 µg/m³ (kept); one differs by 1.56 µg/m³ (excluded).
- Unit conversions applied and documented: ozone ppm to ppb (x1000), temperature in °F, wind in knots (wind direction rows dropped).
- Days with fewer than 2 reporting stations are treated as missing (final panel missingness ≈ 0%).

ETL and database

ETL = Extract, Transform, Load. Extract reads the raw files; Transform applies the cleaning rules above; Load writes clean per-station files, one combined daily table (1,826 days), and a data-quality report. The same data is loaded into an SQLite database (database/atmos_airquality.db) with 4 tables (sites, observations, daily_panel, metadata; 54 sites, 274,467 observations) and demonstrated with SQL queries.

Technology stack (compact)

Area	Tool (Python)
Language / environment	Python 3.13, virtual environment, pytest (9 automated tests)
Data handling	pandas 3.0, numpy 2.5
Database	SQLite 3 (via Python sqlite3)
Statistics	scipy 1.18, statsmodels 0.15 (t-tests, ANOVA, chi-square, OLS inference, Holt-Winters)
Machine learning	scikit-learn 1.9 (Naive Bayes, Random Forest, Linear/Ridge regression, PCA, clustering, metrics)
NLP	scikit-learn TF-IDF + Multinomial Naive Bayes; wordcloud
Visualisation	Plotly 7.1 (interactive), matplotlib (dendrogram, word cloud)
Dashboard	Streamlit 1.64 (12 interactive sections)

Page 3 - Data Analysis & Statistics

Exploratory Data Analysis (EDA)

- Distributions: PM2.5 is right-skewed (skewness 1.61) - most days are moderate, a few days are extreme; a log transform reduces skew to -0.43.
- Anomaly screening: days with |z-score| above 3 are flagged (26 days). The extremes are real events, not errors: the top PM2.5 days are 11 Sep 2024 (40.7 µg/m³, wildfire smoke) and 5 July 2023/2024 (32-35, fireworks).
- Time patterns: monthly means expose the seasonal cycle; scatter plots expose pairwise relationships between pollutants and weather.

Correlation (does one variable move with another?)

Pair	Pearson r	Plain-English reading
Ozone - Temperature	+0.69	Hot days go with high ozone (strongest link in the data)
NO2 - Wind speed	-0.66	Windy days disperse traffic pollution
Ozone - NO2	-0.52	High-NO2 days are cooler/stagnant; ozone needs sun and heat
PM2.5 - NO2	+0.33	Combustion sources raise both
PM2.5 - Wind speed	-0.31	Ventilation lowers particle levels

Both Pearson (linear) and Spearman (rank) correlations are computed; they agree here.

Hypothesis testing

p-value in simple words: the probability of seeing a difference at least this large if there were actually no real difference. A small p-value (below 0.05) means the difference is unlikely to be pure chance - it is "statistically significant". It does NOT mean the difference is large or important, so effect sizes are reported too.

Welch's t-test (implemented): a t-test comparing the means of two groups that does not assume equal variances - the safer default for real data. Each parametric test is cross-checked with a non-parametric version (Mann-Whitney / Kruskal-Wallis).

Question	Test	Result
Is winter PM2.5 different from summer?	Welch t-test	Yes but tiny: 9.9 vs 10.5 µg/m³, p = 0.029, effect size d = -0.15 (small)
Is weekday traffic pollution higher than weekend?	Welch t-test	Yes: NO2 13.7 vs 10.9 ppb, p ≈ 0.0000000001 (effectively zero), d = 0.54 (medium)
Does season explain PM2.5 levels?	One-way ANOVA (4 seasons)	Significant but weak: season explains only ~3% of variance (eta-squared 0.03)
Are ozone exceedance days spread evenly across seasons?	Chi-square	No significant association found (p = 0.39) - an honest null result, kept as found

Lesson emphasised in the project: "significant" is not the same as "large", and a null result is still a result.

Page 4 - Machine Learning

The project answers two questions with models:

- **Classification:** will tomorrow be an ozone exceedance day? (basin daily-max ozone above 70 ppb - yes/no)
- **Regression:** what will the basin's daily-max ozone level be? (a continuous number, in ppb)

Feature	»	Target	»	Training	»	Testing	»	Prediction
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Feature = the input columns (NO2, temperature, wind, calendar). **Target** = the answer to learn. **Training** = the algorithm learns patterns from 2021-2024. **Testing** = it is evaluated on 2025, a full year it has never seen (a strict chronological split - the data is never shuffled, which would leak the future into training). **Prediction** = the trained model applied to the test year.

Leakage guard: because the target comes from ozone, no ozone variable is allowed as an input feature (enforced by an automated test). The model must predict from pollution sources and weather, not from ozone itself.

Models actually implemented, with real 2025-holdout results

Model	Predicts	Inputs	Why used	Key result
Majority baseline	exceedance (class)	-	Honest floor: what "always predict no" scores	PR-AUC 0.118 (floor)
Gaussian Naive Bayes	exceedance (class)	NO2, temp, wind, month, weekend	Fast, simple probabilistic benchmark	PR-AUC 0.639; recall 95%; catches almost every exceedance day, many false alarms
Random Forest (500 trees)	exceedance (class)	same + engineered lags /interactions	Strong nonlinear model; gives feature importances	PR-AUC 0.585; precision 50% - fewer false alarms than NB
Linear Regression (OLS)	ozone level (ppb)	same feature set	Interpretable workhorse: each coefficient shows how one input moves the predicted level	$R^2 = 0.69$; RMSE 6.71 ppb vs 12.05 for a mean guess; temperature (+) and wind (-) significant
Ridge regression	ozone level (ppb)	same	OLS + regularisation to discourage overfitting (strength fixed in advance, not tuned on test)	$R^2 = 0.69$ - identical to OLS
Random Forest (regressor)	ozone level (ppb)	same	Nonlinear check: can trees beat the linear model?	RMSE 6.67 - under 1% better, so the relationship is largely linear

Honest finding: the engineered features (lags, rolling averages, interactions) did **not** improve the 2025 holdout (Random Forest PR-AUC 0.585 with baseline features vs 0.516 with engineered). This is reported as found - results were never tuned on the test year to make them look better.

Page 5 - PCA, Clustering, Association Rules & NLP

PCA - Principal Component Analysis

PCA compresses the five standardised variables into a few combined "components" that keep most of the information. The first two components keep 78% of the variance (46.5% + 31.5%), so each day can be drawn as one point on a 2D map. PCA is used for visualisation and structure, not as a final answer by itself.

Hierarchical clustering - what groups exist among days?

- Ward's method merges similar days step by step; the tree of merges is plotted as a **dendrogram** (implemented in the notebook).
- The number of groups (k) is chosen by the silhouette score over $k = 2..8$, not by eye: $k = 2$ wins (silhouette 0.33).
- **Cluster A (366 days)**: cool, stagnant, combustion-influenced - PM2.5 12.2, NO2 19.7 ppb - ozone exceedance on only 0.8% of days.
- **Cluster B (1,460 days)**: warm and ventilated - ozone 39.5 ppb - exceedance on 17.9% of days.
- Honest framing: silhouette is modest, so the groups overlap; they are patterns in this dataset, not natural laws.

Association rules - which conditions appear together?

Days are binned into simple conditions (e.g. "high temp" = warmest 25% of days). For each rule the project reports **support** (how often the combination occurs), **confidence** (how often the combination really is an exceedance day) and **lift** (how many times more likely than average; 1 = no association). Top findings:

Condition (antecedent)	Confidence	Lift	Reading
high temp & high PM2.5	65%	4.5	hot + hazy days are ~4.5x more likely to be ozone-exceedance days
high temp	47%	3.3	heat alone is the strongest single signal
high PM2.5	26%	1.8	weakly positive on its own
high NO2	3%	0.2	NEGATIVE association: high-NO2 days are cool/stagnant
low wind	1%	0.1	NEGATIVE: calm days here are cool-season inversion days, not ozone days

Association is NOT causation. These rules say which conditions co-occur in this dataset. They do not prove that any factor causes ozone exceedance.

NLP - text intelligence (secondary dataset)

- The sensor data has no text column, so a clearly separated real corpus is used: 1,200 US Federal Register EPA documents about air quality (public domain).
- Text is turned into numbers with TF-IDF (how distinctive each word is per document class).
- **Multinomial Naive Bayes** classifies Rule vs Notice documents: **95.3% accuracy, F1 0.949** (trained on 900, tested on 300 held-out documents).
- A **word cloud** visualises the most frequent terms after removing common English stopwords.

Page 6 - Forecasting & Dashboard

What is being forecast

The weekly average PM2.5 of the basin. The series is split in time: models learn from 2021-2024 and must forecast the whole of 2025 (about 53 weeks) in one multi-step shot - no peeking at the test year.

Methods (statsmodels)

- **Holt's method** - extrapolates the trend (damped so it does not run away).
- **Holt-Winters** - trend + yearly seasonality (52-week cycle).
- **Seasonal-naive baseline** - "same week last year"; the honesty floor every forecast must beat.

Actual result (2025 holdout)

Model	RMSE ($\mu\text{g}/\text{m}^3$)	MAE	Verdict
Holt-Winters (additive)	3.33	2.73	Best - real but modest skill
Seasonal-naive baseline	4.17	3.02	Beaten by Holt-Winters
Holt (damped trend)	4.76	4.30	Worse - a plain trend is not enough

Limitations: average error is about one-third of the mean level (MAPE ~32%); wildfire episodes cannot be anticipated from history alone, so forecast accuracy is genuinely bounded. This is a statistical demonstration of the syllabus topic, not an operational forecast service.

The Streamlit dashboard (12 interactive sections, Plotly charts)

Section	What it shows
1. Executive Overview	Key numbers (1,826 days, 265 exceedance days, 54 monitors) and rolling trends
2. Data Quality	Cleaning report; how many monitors reported each day; missing values
3. Exploratory Analysis	Distributions, monthly patterns, scatter matrix, record days
4. Statistical Intelligence	Confidence intervals, the four hypothesis tests, z-score anomaly chart
5. Risk Screening (ML performance)	Model comparison, confusion matrices, a what-if slider demo
6. Feature Importance	Which inputs the Random Forest relied on (with a no-causation caveat)
7. Regression (O3 level)	OLS R^2 /RMSE, observed-vs-estimated scatter, full coefficient table
8. PCA & Clustering	Explained variance, day-map in PC space, cluster profiles
9. Association Analysis	Lift chart of condition rules (co-occurrence, not causation)
10. Text Intelligence (NLP)	Naive Bayes metrics and word cloud on the EPA corpus
11. Forecasting	2025 holdout forecast vs actual, model error table
12. Key Findings	The nine headline results, each with its supporting number

The dashboard runs offline on historical data and is explicitly labelled "not a live monitoring system".

Page 7 - Key Findings & Limitations

Major findings (each verified from executed project code)

#	Finding	Supporting result	One-line explanation
1	Heat is the main ozone driver	O3-temperature correlation +0.69; "high temp" rule lift 4.5	Hot, sunny days cook traffic emissions into ozone
2	Weekday traffic signal is real	NO2 13.7 (weekday) vs 10.9 ppb (weekend), $p \approx 0$, $d = 0.54$	Rush-hour pollution drops on weekends
3	Days come in two regimes	Ward clustering, $k = 2$; exceedance share 0.8% vs 17.9%	Cool/stagnant/combustion days vs warm/ventilated days
4	Ozone level is largely predictable from weather	OLS R^2 0.69 (RMSE 6.7 vs 12.1 ppb baseline)	A simple linear model explains ~69% of day-to-day variance
5	Exceedance screening works, with trade-offs	Naive Bayes PR-AUC 0.639 vs 0.118 floor; recall 95%, precision 38%	Catches nearly every bad day but raises many false alarms
6	Forecasting beats the naive baseline	Holt-Winters RMSE 3.33 vs 4.17 $\mu\text{g}/\text{m}^3$	Seasonal statistics add genuine skill
7	Real events appear in the data	Top PM2.5 days: 11 Sep 2024 wildfire (40.7), 5 Jul 2023/2024 fireworks	Extremes are genuine episodes, not data errors
8	More complexity did not help	RF PR-AUC 0.585 (baseline) vs 0.516 (engineered features)	Simple features were enough; reported honestly

Limitations (stated openly)

- **Data:** basin-average values can hide local hotspots; monitor coverage and methods changed slightly over the years (documented in the quality report).
- **Geography:** one air basin in Southern California - findings may not transfer to other cities or climates.
- **Missing data:** negligible in the final daily panel ($\approx 0\%$ after the 2-station rule), but individual stations have gaps; PM2.5 has fewer stations (20) than ozone (44).
- **Exceptional events:** wildfire smoke (e.g. September 2024) inflates PM2.5 extremes; such days are kept and flagged, not removed.
- **Statistics/ML:** consecutive days are correlated, so effective sample sizes are smaller than they look and p-values are optimistic (Durbin-Watson 0.92 in the regression); the classifier has limited precision; no model proves causation.
- **Forecasting:** ~32% MAPE; wildfire episodes are inherently unpredictable from past data.
- **NLP:** secondary corpus only (US federal air-quality documents); it demonstrates the text-mining syllabus topic and is separate from the sensor analysis.

Page 8 - How I Can Explain It to My Ma'am

Short answers I can memorise

Question	My answer
1. What is your project?	"My project is called Atmos. It is a complete data science study of urban air pollution in the Los Angeles area, using five years of official US EPA monitoring data, and it ends with an interactive dashboard."
2. Why did you choose this project?	"Air pollution affects everyone's health, and I wanted a project with real public data where honest analysis matters - not a toy dataset. It also lets me demonstrate the full syllabus: statistics, machine learning, clustering, NLP, forecasting and visualisation on one real problem."
3. What dataset did you use?	"The US EPA Air Quality System - daily readings of PM2.5, ozone, NO2, temperature and wind from 54 monitors around Los Angeles, from 2021 to 2025 - about 274,000 observations. For the NLP part I used 1,200 real EPA documents from the Federal Register."
4. What did you do with the data?	"I built an ETL pipeline that cleans and combines everything into one daily table, loaded it into SQLite, explored it with statistics, ran hypothesis tests, trained classification and regression models with a strict time-based split, clustered the days, mined association rules, classified text with Naive Bayes, forecast PM2.5 with Holt-Winters, and put all of it on a Streamlit dashboard."
5. Which ML algorithms did you use?	"Naive Bayes and Random Forest to predict ozone exceedance days, and Linear Regression - plus Ridge and a Random Forest regressor as references - to estimate the ozone level. Naive Bayes caught 95% of exceedance days, and the linear model explained 69% of the ozone variance."
6. What did you find?	"Three things stand out: hot days drive ozone (correlation +0.69); weekday traffic pollution is clearly higher than weekend; and days split into two regimes - warm ventilated days with ozone problems versus cool stagnant days with particle and NO2 problems. I also report honestly where complexity did not help."
7. What is the dashboard?	"A 12-section Streamlit web app with interactive Plotly charts - overview, data quality, statistics, model performance, regression, clustering, associations, NLP, forecasting and key findings. It runs offline on the historical data and is clearly labelled as not a live system."
8. What are the limitations?	"It covers one air basin, basin averages can hide local hotspots, wildfire episodes inflate the extremes, consecutive days are correlated so p-values are optimistic, and all findings are associations - not causation."

Complete 30-45 second explanation (say it in your own words)

"My project, Atmos, turns five years of official EPA air-quality data from the Los Angeles area into insight. I built an ETL pipeline that cleans 274,000 observations into one daily table and a SQLite database. I explored the data with descriptive statistics and hypothesis tests - for example proving the weekday traffic signal in NO2. I trained machine-learning models with a strict chronological split: Naive Bayes and Random Forest to flag ozone exceedance days - catching 95% of them - and Linear Regression to estimate the ozone level, which explains 69% of its variance, mostly through temperature and wind. I also clustered days into two regimes, mined association rules, classified real EPA documents with Naive Bayes at 95% accuracy, and forecast PM2.5 with Holt-Winters, beating the naive baseline. Everything is presented in a 12-section Streamlit dashboard, with limitations stated openly: one region, associations not causation, and wildfire effects."

Deliverables: 13 Python modules in src/, 6 executed notebooks, SQLite database, pytest suite (9 tests), full documentation (README, methodology, leakage audit, limitations) and the interactive dashboard.